Thesis Talk

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Overview of Topics

- Context
- Theoretical Introduction (sorry)
- Overview of Models
- Results of Cohesive Zone Model
- Disucussion

What am I doing and why is this relevant?

- Hang out in my office and have coffee (please come by)
- Trying to make an accurate model for crack propagation in snow
- Cracks in snow on large scales can lead to avalanche release
- Average of 24 deaths by avalanche in Switzerland per year, multiple hundred non-fatal accidents (SLF, 2025)
- Understanding relevant because avalanches endanger human life and infrastructure (Schweizer et al., 2003)
- **Ultimate Goal:** determining whether slab breakoff can be seen in DAS data and how this would look

Words I will propably say a lot explained I

- Slab: Part of the snowpack which breaks off and slides downhill as a cohesive unit (the avalanche itself, I'm looking at dry-snow slab avalanches) (Schweizer et al., 2003)
- Weak layer: Layer with different mechanical properties than the rest of the snow, cracks usually propagate in this layer because it's weaker than the rest of the snowpack (Schweizer et al., 2003)
- Mode I, Mode II and Mode III cracks: cracks opening as a response to stress in different directions:
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Bibliography I

Schweizer, J., Bruce Jamieson, J., & Schneebeli, M. (2003).Snow avalanche formation. *Reviews of Geophysics*, *41*(4). https://doi.org/10.1029/2002RG000123

SLF. (2025). Avalanche Victims since 1936 - Long Term Statistics.

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